

Chapter 5: Results and Evaluation

This chapter assesses the strengths and weaknesses of the BRAVE-based resynthesis pipeline when it is used in real-time situations. The study mainly concentrates on how stable the latent representations are, the behavior of adversarial training, and the ability to generalize out-of-distribution (OOD) across the guitar and drum resynthesis tasks. We examine the results both quantitatively and perceptually in an attempt to find the main obstacles that prevent efficient streaming neural audio resynthesis.

The focus of this research is divided into two main projects: a six iteration development for guitar resynthesis and an additional test for drum-kit resynthesis. As described in the methodology, the results of the pipeline were measured against certain diagnostic criteria for encoder separation (pairwise latent difference greater than 0.5), decoder sensitivity (pairwise output difference greater than 0.01), and energy conservation (output level within $[0.05, 0.3]$) (see §3.4). Besides that, the in-distribution reconstruction was checked by a Short-Time Fourier Transform (STFT) magnitude correlation rule of greater than 0.3 (see §3.4). All live experiments were done in a Pure Data environment with the use of the `nn~` external and an estimated end-to-end delay of about 7 ms (see §3.1).

5.1 Guitar, Six-Iteration Arc

The guitar example showed how to gradually improve the training procedure piece by piece, initially starting with very simple settings then going all the way to using the RAVE v2 model, which is well known in the community. Each step was performed by running the diagnostic Python scripts mentioned in Chapter 4 to precisely identify the problem areas.

5.1.1 Iterations v1, v2: Latent Space Failures

The first version (guitar_v1) was based on the `c128_r10.gin` file without any changes (refer to §4.2). It was found that this run was affected by the posterior collapse issue, which makes the latent space unusable since, in the beginning, the Kullback-Leibler (KL) component is too strong and pushes the solution too much toward the prior (Bowman et al., 2016). Using TensorBoard, it was found that the regularization (KL) term value decreased from 0.5 to 0.25 over the time of training. Moreover, although the encoder was able to distinguish between inputs ($\text{diff} = 1.25$), the decoder simply discarded the latent information, resulting in a forward difference of 0.006, which was far less than the 0.01 level required. Hence, it was considered that a log-space β warmup was needed for the model to first learn significant reconstructions before being regularized (see §3.2).

On the other hand, in the next run (guitar_v2) the β warmup technique was carried out and posterior collapse was no longer a problem. The model, however, did suffer a partial collapse because the latent space was too large. The `LATENT_SIZE` was 128 and as a result, the model was only able to use about 0.005 nats per dimension. The tests revealed a small encoder difference (0.79) and a decoder that was not reacting at all ($\text{forward diff} = 0.003$). The output amplitude of 0.012 was roughly 60 times lower than the input signal. This kind of problem implied that the 128-dimensional latent vector was too large for the ~5.4-hour corpus that was available at that stage, and so a reduction down to 16 dimensions was necessary (see §3.2).

5.1.2 Iterations v3-v5: Adversarial Instabilities

The 3rd and 4th iterations focused on the data composition and its influence on the model; the latent size was decreased to 16. Having been trained on 5.79 hours of mixed data, the guitar_v3 model exhibited a very weak response to forward differences, which at the highest reached 0.042. However, the model experienced Phase 2 generator collapse, meaning that the custom discriminator has been the winner of the adversarial game (see §4.4).

The guitar_v4 attempt was to use a cleaner, only microphone recording data set (3.05 h vs 5.79 h for v3). Quite paradoxically, the decrease of the data amount led to more failure in the adversarial mode as shown by TensorBoard scalars. The discriminator clearly dominated: the `pred_real` reached 3.34 while `pred_fake` went down to -2.31, which produced a logit gap of 5.65. The decoder outputted a silence amplitude larger than the signal amplitude under these circumstances.

A focused experiment for making the adversarial game more balanced was guitar_v5 where the discriminator's capability was halved (see §4.2). This led to the opposite error: the failure of the discriminator. The `pred_real` logit dropped to -0.90, which means the discriminator was giving negative marks even to real audio, and the gap became 0.80. These results, which are in short form in Table 5.1, mean that merely scaling parameters of a custom architecture will not be able to solve the issue of which one dominates/collapses.

Iteration	Configuration Main Change	Failure Mode	Diagnostic Signature
v1	Baseline c128_r10.gin	Posterior collapse	KL fell 0.5 $\rightarrow$ 0.25; forward diff 0.006
v2	β warmup applied	Latent underutilisation	KL/dim $\approx$ 0.005 nats; forward diff 0.003
v3	LATENT_SIZE $\rightarrow$ 16	Phase 2 collapse	Forward diff peaks at 0.042 then degrades
v4	Mic-only data (3.05 h)	Mode collapse	Gap = 5.65; silence amp > signal amp
v5	Halved discriminator	Discriminator collapse	<code>pred_real</code> = -0.90; gap = 0.80
v6	Official RAVE v2 base	Working autoencoder	STFT correlation passes 0.3; GAN gap reversal

Table 5.1: Summary of failure modes across the guitar development iterations.

5.1.3 Iteration v6: Working Autoencoder

The final release (guitar_v6) was based on the RAVE v2 official architecture and `causal.gin` overrides (see §4.2). This setup was the first to produce a working autoencoder responsive to input data drawn from the same distribution. In Phase 2, the GAN logit gap was seen to reverse for the first time ever in the series, suggesting that an adversarial equilibrium was reached. Listening tests suggested that reconstruction quality peaked at epoch 1935 (mid-Phase-2), which is why this checkpoint was selected as canonical, with the final epoch (2511) showing slightly degraded performance.

However, the model was not able to reproduce the signal after receiving voice input that was not in the training distribution (OOD). The encoder was capable of differentiating the vocal signals (latent differences ranged from 1.27 to 6.89), but the decoder produced a practically uniform, low-energy output. This suggests that the encoder can encode voice into latent space, but the resulting trajectories are outside of the guitar decoder's trained manifold.

5.2 Drums

The evaluation of the drum-kit was an attempt to port the BRAVE streaming-causal framework of Caspe et al. (2025) to the 10.86-hour Groove corpus (see §3.3).

The initial attempt at drum-kit (drums_v1), by using `c16_r10_beta_fixed.gin` config, resulted in a forward difference of 0.0045. This was lower than the 0.01 threshold set for the tonal responsiveness but listening to the

model's output indicated that there was a partial transient that the model was able to respond to. In Pure Data, voice consonants and beatbox transients seemed to give rise to recognisable drum-like timbres. However, strictly speaking, the numerical diagnostic tells us that a "working voice-to-drums" claim cannot be substantiated by solely spectral evidence.

The second line of work on drums (drums_v2) followed the v2.gin towards the successful application to the Groove corpus and was training for about 95 hours (≈ 2.28 M steps). Phase 1 ended with good validation scores (val ≈ 4.1 , mb_spec ≈ 3.7). Yet, Phase 2 saw a steady increase in the GAN gap from 9.8 to 20.3 over 800,000 steps, while the validation loss was basically stagnant around 7.5. Training was stopped even before the scheduled time as the discriminator kept gaining the upper hand despite the update_discriminator_every=4 feature in the v2 architecture, which was supposed to be a safeguard. This means that the v2.gin recipe did not really work for the drums dataset when subjected to the exact conditions of training used in this experiment, unlike in the guitar dataset.

5.3 External Baseline: IRCAM Percussion

To see if the OOD problems from iterations v6 and drums_v1 are the cases for the BRAVE architecture in general, a side-by-side evaluation was done using the official ACIDS-IRCAM percussion model (71 MB TorchScript). It was given the same synthetic-input test as described in §3.4 and the IRCAM model came out with a forward difference (silence vs. noise) of 1.75, which means that the official model is $1.75/0.0045 \approx 390$ times more responsive than the drums_v1 iteration.

The IRCAM model respected silence (0.001 output for silent input) and, in informal Pure Data listening tests with voice input, produced audibly modulated percussive output. This indicates that a well-trained BRAVE model with higher-quality or more voluminous data can overcome the manifold-mapping bottleneck. This comparison supports the conclusion that the performance-limiting factors of the present models are training recipe and data scale rather than fundamental architectural constraints of BRAVE.

5.4 Failure-Mode Taxonomy

The outcomes of the ten-week training period could be summarised into five chief failure modes, which are significant for streaming VAE resynthesis. These modes, which were discovered through the Python diagnostics that were done in Chapter 4, serve as guides to the stabilisation of similar real-time models.

1. **Posterior Collapse (v1):** Identifiable by a KL curve that keeps falling and a decoder that no longer watches the latent space. A mistake could be made here by trying to put KL pressure before reconstruction has been learned.
2. **Latent Underutilization (v2):** It can also be identified by very low nats-per-dimension (≈ 0.005) and a near-silent decoder. It happens when the bottleneck capacity is larger than the information density of the dataset.
3. **Discriminator Dominance / Generator Mode Collapse (v3, v4, drums_v2):** Should be a cause of concern when a discriminator logit gap continues to expand (usually > 5) and, ultimately, the silence amplitude is higher than the signal amplitude. The trigger is when the default custom discriminator is out-pacing the generator during the Phase 2.
4. **Discriminator Collapse (v5):** The inverse failure—heuristic discriminator weakening—pushed pred_real to negative values and narrowed the logit gap to 0.80. Confirmed that no simple capacity-scaling resolves the dominance/collapse axis.
5. **OOD Decoder Behavior (v6, drums_v1):** It is a major defect at the structural level in which the encoder can distinguish a non-instrument input (e.g., voice), while the decoder is not able to produce it because the latent resulting coordinates lie outside the trained manifold.

As the IRCAM baseline comparison confirms, these should be regarded as training-recipe failures. Addressing them in future work will require more homogeneous datasets and possibly explicit pitch-conditioning to close the cross-modal gap.

Having characterised the training instabilities and documented a working guitar autoencoder in guitar_v6, the final chapter summarises the main technical findings and proposes concrete steps for improving voice-driven control.